CRED-nf checklist summary

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Manuscript title: Effectiveness of Continuous-Intermittent Hybrid Neurofeedback on Alpha Regulation and Discrimination
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Item No.	Checklist item	Manuscript Details
Pre-experiment		
1a	Pre-register experimental protocol and planned analyses	Methods: "The study was preregistered on the Open Science Framework (OSF): https://osf.io/2qg5j."
1b	Justify sample size	Methods: "Sample size was determined a priori: a power analysis indicated that 27 participants would be required to achieve 80% power for detection of a difference between conditions. We anticipated a 10% exclusion rate, therefore recruitment targeted a sample size above 30."
2a	Employ control group(s) or control condition(s)	Partially. Discussion: "[...] the experiment did not include a no-feedback control condition". Methods: "The study used a within-subjects design with two feedback conditions: hybrid (continuous sonification plus intermittent reward tones) and intermittent (reward tones only). The intermittent condition served as the comparison condition for the added continuous sonification."
2b	When leveraging experimental designs where a double-blind is possible, use a double-blind	*NA: A double-blind was not appropriate for this experiment*
2c	Blind those who rate the outcomes	Methods: "Trial outcomes (success in the regulation task, correctness in the discrimination task) were determined automatically by the online pipeline according to the threshold criterion, without human rating."
	Blind those who analyse the data	*Those who analysed the data were not blind to group assignment*
2d	Examine to what extent participants and experimenters remain blinded	*NA: There was only one participant group*
2e	In clinical efficacy studies, employ a standard-of-care intervention group as a benchmark for improvement	*NA: This is not a clinical efficacy study*

Control measures		
3a	Collect data on psychosocial factors	*Psychosocial factors were not measured*
3b	Report whether participants were provided with a strategy	Methods: "They were given a suggested mental strategy for each direction: visual imagery and active thinking to decrease alpha amplitude, and relaxing and clearing their mind to increase it."
3c	Report the strategies participants used	*The strategies participants used were not recorded or not reported in the manuscript*
3d	Report methods used for online-data processing and artifact correction	Methods: "The ongoing absolute alpha amplitude (AAA) was estimated using a ring buffer structure, creating a 4-second sliding window with 90% overlap, updated at a sampling rate of 2.5 Hz. [...] On each update, the entire buffer was filtered between 8-12 Hz using a bandpass-filter, then the instantaneous amplitude envelope was calculated and averaged to obtain an estimate of the AAA during the last 4 seconds. The online pipeline introduced an estimated total delay of approximately 3.08 seconds [...]. See Supplementary material C for more ring buffer and online analysis details."
3e	Report condition and group effects for artifacts	*Condition and group effects for artifacts were not measured, or not reported in the manuscript*
Feedback specifications		
4a	Report how the online-feature extraction was defined	Methods: "Both the window duration and the use of absolute rather than relative amplitude were chosen following the recommendation by Frederick (2012) to improve discrimination performance. The buffer contained data exclusively from the O2 electrode, as alpha activity is typically stronger at occipital sites (Frederick, 2012; Nunez, 2005), and because using a single fixed channel provided a consistent and stable source across participants."
4b	Report and justify the reinforcement schedule	Methods: "Both tasks under both conditions included intermittent feedback in the form of a reward sound contingent on successful trials (regulation) and correct answers (discrimination). The reward sound chosen was "Tada.wav" (Microsoft Corporation, 1992), as in Frederick et al. (2016)."

4c	Report the feedback modality and content	Methods: "Both tasks under both conditions included intermittent feedback in the form of a reward sound contingent on successful trials (regulation) and correct answers (discrimination). The reward sound chosen was "Tada.wav" (Microsoft Corporation, 1992), as in Frederick et al. (2016). In the hybrid condition, the regulation task additionally included continuous auditory feedback: a tone whose volume tracked ongoing alpha amplitude in near-real time." "The continuous auditory feedback was implemented by mapping the ongoing AAA onto the volume of a continuous bandpass-filtered (50-250 Hz) red noise tone, generated using Purr-Data (Gräf, 2024). [...] higher alpha amplitude corresponding to higher sound volume. [...] See Supplementary Material E for more continuous feedback mapping details."
4d	Collect and report all brain activity variable(s) and/or contrasts used for feedback, as displayed to experimental participants	Methods: "This estimate [Absolute Alpha Amplitude] was then used for the alpha state evaluation and the feedback calculation. See Supplementary material C for more ring buffer and online analysis details." "The continuous auditory feedback was implemented by mapping the ongoing AAA onto the volume of a continuous bandpass-filtered (50-250 Hz) red noise tone, generated using Purr-Data (Gräf, 2024). "
4e	Report the hardware and software used	Methods: "The experiment was programmed in MATLAB R2022b (The MathWorks, Inc., 2022a) using the Psychtoolbox v3.0.18 extension (Kleiner et al., 2007), EEGLAB 2025.0.0 (Delorme & Makeig, 2004) and Lab Streaming Layer (LSL) (Kothe et al., 2025). Continuous feedback was produced in Purr Data (Gräf, 2024). Participants' EEG were recorded using a SMARTING Pro mobile EEG system (mBrainTrain, 2025) with 32 electrodes arranged according to the international 10-20 system."
Outcome measures - brain		
5a	Report neurofeedback regulation success based on the feedback signal	Methods: "Following Frederick et al. (2016), we applied their performance criteria to classify participants who demonstrated successful NF training. For the regulation task, the criterion was defined as at least a 10% alpha amplitude difference between the AAA during the up- and during the down-regulation trials (see Supplementary Material H)."
5b	Plot within-session and between-session regulation blocks of feedback variable(s), as well as pre-to-post resting baselines or contrasts	Supplementary Material I: "Figure S8: Normalized AAA across the session. AAA was z-scored within each participant. The first violin plot shows baseline AAA normalized values, followed by Block 1 to 4. Gray lines connect each participant's values across blocks; orange lines mark participants whose AAA in that block exceeded their own baseline value."
5c	Statistically compare the experimental condition/group to the control condition(s)/group(s) (not only each group to baseline measures)	Results: "A paired t-test indicated no significant difference in amplitude difference between the hybrid and intermittent conditions, $t(26) = 1.039$, $p = .851$, Cohen's $d = 0.200$. "
Outcome measures - behaviour		

6a	Include measures of clinical or behavioural significance, defined a priori, and describe whether they were reached	*The manuscript does not include measures of clinical or behavioural significance*
6b	Run correlational analyses between regulation success and behavioural outcomes	Results: "On average, regulation task performance was slightly higher in the hybrid condition (M = 60.419%, SD = 11.401) than in the intermittent condition (M = 58.914%, SD = 12.186), see Figure 4a. However, the paired t-test indicated no significant effect of the hybrid feedback relative to only the intermittent feedback, t(26) = 0.551, p = .585, Cohen's d = 0.106. Similarly, discrimination performance was slightly higher in the hybrid condition (M = 49.395%, SD = 13.211) than in the intermittent condition (M = 47.618%, SD = 12.773), see Figure 4b. However, the paired t-test indicated no significant effect of prior exposure to the hybrid feedback relative to only the intermittent feedback, t(26) = 0.548, p = .588, Cohen's d = 0.105."
Data storage		
7a	Upload all materials, analysis scripts, code, and raw data used for analyses, as well as final values, to an open access data repository, when feasible	Methods: "Experiment code, analysis scripts, and study materials are available at github.com/alexisbaxman/alpha-neurofeedback-hybrid. Raw participant data are not publicly available, as participant consent did not cover public data sharing."